

DAY-5 Hands-On Activities – Keerthivasan R

Problem 1: Greeting Generator (Level-1)

Scenario

A simple webpage should greet users based on the name they enter. This helps beginners understand how JavaScript functions accept parameters and how variables behave inside functions.

Requirements

- Create an input box to enter a user's name.
- Create a button labeled "Generate Greeting".
- When the button is clicked:
 - A JavaScript function should accept the name as a parameter.
 - Display a greeting message like:
"Hello, Rahul! Welcome to our website."
- Display the greeting inside a `<p>` element.

Technical Constraints

- Use only vanilla JavaScript (no libraries).
- Use function keyword (not arrow functions for this task).
- Use `document.getElementById()` for DOM manipulation.
- Variable for the greeting message must be declared inside the function.

Learning Outcome

You should be able to:

- Understand how functions receive input using parameters.
- Learn local variable scope.
- Manipulate webpage elements using the DOM.

CODE:

```
<!DOCTYPE html>
<html>

<head>
  <!-- Title shown in browser tab -->
  <title>Greeting Generator</title>
</head>
```

```
<body>

  <h2>Greeting Generator</h2>

  <!-- Input field to enter username -->
  <input type="text" id="username" placeholder="Enter your name">

  <!-- Button calls generateGreeting() when clicked -->
  <button onclick="generateGreeting()">Generate Greeting</button>

  <!-- Paragraph where greeting will be displayed -->
  <p id="greeting"></p>

  <script>

    // Function triggered when button is clicked
    function generateGreeting() {

      // Get value from input field
      var name = document.getElementById("username").value;

      // Local variable (only accessible inside this function)
      var message = "Hello, " + name + "! Welcome to our website.";

      // Display greeting inside paragraph
      document.getElementById("greeting").innerHTML = message;
    }

  </script>

</body>

</html>
```

OUTPUT:

Greeting Generator

Hello, Keerthivasan! Welcome to our website.

Problem 2: Simple Object-Based User Info Display (Level-1)

Scenario

A webpage needs to display basic user information stored inside a JavaScript object.

✚ Requirements

Create a JavaScript object user with properties:

- name
- age
- city

Create a function displayUserInfo(userObj):

- Accepts the object as a parameter.
- Displays user details in separate <p> elements.

A button click should trigger the function.

🔧 Technical Constraints

- Object properties must be accessed using dot notation.
- Function should not use global variables.
- DOM elements should be updated dynamically.

🎯 Learning Outcome

You will be able to:

- Understand JavaScript objects.
- Pass objects to functions.
- Display object data dynamically on a webpage.

CODE:

```
<!DOCTYPE html>
<html>

<head>
  <title>User Info Display</title>
</head>

<body>

  <h2>User Information</h2>

  <!-- Button triggers showUser() -->
  <button onclick="showUser()">Display User Info</button>

  <!-- Paragraphs to display user details -->
  <p id="name"></p>
```

```
<p id="age"></p>
<p id="city"></p>

<script>

    // JavaScript object (stores user data)
    var user = {
        name: "Rahul",
        age: 22,
        city: "Delhi"
    };

    // Function that accepts an object as parameter
    function displayUserInfo(userObj) {

        // Access object properties using dot notation
        document.getElementById("name").innerHTML = "Name: " + userObj.name;
        document.getElementById("age").innerHTML = "Age: " + userObj.age;
        document.getElementById("city").innerHTML = "City: " + userObj.city;
    }

    // Calls displayUserInfo and passes global user object
    function showUser() {
        displayUserInfo(user);
    }

</script>

</body>

</html>
```

OUTPUT:

User Information

Display User Info

Name: Rahul

Age: 22

City: Delhi

Problem 3: Counter App with Scope Control (Level-2)

Scenario

Build a counter application where users can increment and reset a value. This problem focuses on variable scope and DOM manipulation.

📌 Requirements

Display a counter value starting from 0.

Create two buttons:

- **Increment**
- **Reset**

Use:

- A global variable to store counter value.
- A function `incrementCounter(step)` that:
 - Accepts step value as a parameter.
 - Updates the counter.

Reset button should reset the counter to 0.

🔧 Technical Constraints

- Counter value must be maintained outside the function (global scope).
- DOM updates must happen inside functions only.
- No inline JavaScript in HTML.

🎯 Learning Outcome

Learners should be able to:

- Understand variable scope (global vs local).
- Apply function parameters in real scenarios.
- Control UI behavior using JavaScript functions.

CODE:

```
<!DOCTYPE html>
<html>

<head>
  <title>Counter App</title>
</head>

<body>
```

```
<h2>Counter App</h2>

<!-- Displays counter value -->
<h3 id="counter">0</h3>

<!-- Buttons -->
<button id="incrementBtn">Increment</button>
<button id="resetBtn">Reset</button>

<script>

    // Global variable (accessible throughout script)
    var counterValue = 0;

    // Event listener for increment button
    document.getElementById("incrementBtn").addEventListener("click", function () {
        incrementCounter(1); // Passing step value as argument
    });

    // Event listener for reset button
    document.getElementById("resetBtn").addEventListener("click", function () {
        resetCounter();
    });

    // Function with parameter
    function incrementCounter(step) {

        // Update global variable
        counterValue = counterValue + step;

        // Update UI
        document.getElementById("counter").innerHTML = counterValue;
    }

    function resetCounter() {

        // Reset value to 0
        counterValue = 0;

        // Update UI
        document.getElementById("counter").innerHTML = counterValue;
    }

</script>

</body>

</html>
```

OUTPUT:

Counter App

5

Increment

Reset

Problem 4: Dynamic Student Profile Manager (Level-2)

Scenario

Create a mini student profile system where details are stored in an object and displayed dynamically using DOM manipulation.

📌 Requirements

Create a student object with:

- name
- rollNo
- marks

Create a function `updateStudentProfile(studentObj)`:

- Accepts the object as a parameter.
- Displays details in a styled `<div>`.

Add another function `updateMarks(newMarks)`:

- Updates marks and refreshes the display.

🔧 Technical Constraints

- Student object must be declared in global scope.
- Functions should update object values using parameters.
- DOM should update without page refresh.

🎯 Learning Outcome

You will be able to:

- Deep understanding of object manipulation.
- Function interaction with shared data.
- Real-world use of DOM updates and scope handling.

CODE:

```
<!DOCTYPE html>
<html>

<head>
  <title>Student Profile Manager</title>
</head>

<body>

  <h2>Student Profile Manager</h2>

  <!-- Button to display profile -->
  <button onclick="updateStudentProfile(student)">Show Profile</button>
  <br>

  <!-- Input to update marks -->
  <input type="number" id="newMarks" placeholder="Enter new marks">

  <!-- Button to change marks -->
  <button onclick="changeMarks()">Update Marks</button>

  <!-- Profile display box -->
  <div id="profileBox"></div>

  <script>

    // Global object
    var student = {
      name: "Amit",
      rollNo: 101,
      marks: 85
    };

    // Function to display student data
    function updateStudentProfile(studentObj) {

      var content =
        "<p><strong>Name:</strong> " + studentObj.name + "</p>" +
        "<p><strong>Roll No:</strong> " + studentObj.rollNo + "</p>" +
        "<p><strong>Marks:</strong> " + studentObj.marks + "</p>";

      document.getElementById("profileBox").innerHTML = content;
    }

    // Function to update marks
    function updateMarks(newMarks) {

      // Modify global object property
      student.marks = newMarks;

      // Refresh UI after update
      updateStudentProfile(student);
    }
  </script>
</body>
</html>
```

```
// Get value from input and pass as argument
function changeMarks() {

    var marksValue = document.getElementById("newMarks").value;

    if (marksValue === "") {
        alert("Please enter marks");
        return;
    }

    updateMarks(marksValue);
}

// Automatically display profile when page loads
window.onload = function () {
    updateStudentProfile(student);
};

</script>

</body>

</html>
```

OUTPUT:

Student Profile Manager

Name: Amit

Roll No: 101

Marks: 85